

IEEE Software Engineering-Related Standards

台科大資管系軟體工程與管理研究中心整理

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序 號	標準編號	標準名稱
1	730-2026	IEEE Standard for Software Quality Assurance Processes Abstract : Requirements for initiating, planning, controlling, and executing the software quality assurance (SQA) processes of a software development or maintenance project are established in this standard. This standard is harmonized with the software life cycle processes of ISO/IEC/IEEE 12207:2017, IEEE Std 2675™-2021, and the information content requirements of ISO/IEC/IEEE 15289:2019.
2	1012 -2024	IEEE Standard for System, Software, and Hardware Verification and Validation Abstract : The Verification and Validation (V&V) processes are used to determine whether the development products of a given activity conform to the requirements of that activity and whether the product satisfies its intended use and user needs. V&V life cycle process requirements are specified for different integrity levels. The scope of V&V processes encompasses systems, software, and hardware, and it includes their interfaces. Systems, software, and hardware being developed, maintained, or reused [legacy, commercial off-the-shelf (COTS), non-developmental items] can be the system of interest to which this standard is applied. The term software also includes firmware and microcode, and each of the terms system, software, and hardware includes related information or documentation. V&V processes include the analysis, evaluation, review, inspection, assessment, and testing of products.
3	1062 -2019	IEEE Recommended Practice for Software Acquisition Abstract : A set of useful quality considerations that can be selected and applied during one or more steps in a software acquisition process is described in this recommended practice. The recommended practices can be applied to software that runs on any computer system regardless of the size, complexity, or criticality of the software. The software supply chain may include integration of off-the-shelf (OTS), custom, or free and open source software (FOSS). Each organization or individual

		using this recommended practice will need to identify the specific quality and activities that need to be included within its acquisition process.
4	1636-2018	IEEE Standard for Software Interface for Maintenance Information Collection and Analysis (SIMICA) Abstract : Promoting and facilitating interoperability between components of automatic test systems where test results and/or maintenance actions need to be shared is addressed in this standard. The standard defines the common elements between both test results data and maintenance action data. The common schema becomes a class of information that shall be used within the SIMICA family of standards.
5	2841-2022	IEEE Recommended Practice for Framework and Process for Deep Learning Evaluation Abstract : The recommendations on evaluating and improving algorithm reliability for shortening the development cycle of deep learning algorithms and improving the quality of software systems based on deep learning algorithms are defined in this document. An assessment index system and corresponding assessment process are specified in this document. Scope: This recommended practice stipulates requirements for deep learning algorithms assessment and covers the following aspects: 1. Assessment index system 2. Assessment process 3. Assessment of such stages as demand, design, implementation, and operation
6	2941.2-2023	IEEE Standard for Application Programming Interfaces (APIs) for Deep Learning (DL) Inference Engines Abstract: A set of application programming interfaces (APIs) that is aimed at breaking down the barriers between different deep learning (DL) inference engines and applications is defined in this standard. The APIs are comprised of functional interfaces including parameter reading, model compilation optimization, operator registration, thread management, input/output data acquisition, inference instance creation, and inference execution. Scope: This standard defines a set of application programming interfaces (APIs) for use with different deep learning (DL) inference engines. The interfaces include parameter reading, model compilation optimization, operator registration, thread management, input/output data acquisition, inference instance creation, and inference execution. Purpose: This standard defines efficient APIs for model management, device management, input/output, and inference that break down the barriers between different DL inference engines and applications. The target DL applications include natural language processing, image classification, and object detection.
7	12207-2026	ISO/IEC/IEEE International Standard - Systems and software engineering -- Software life cycle processes Abstract : This document establishes a common process framework for describing

		the full life cycle of software systems from conception through retirement. It applies to the acquisition or development of software products and services whether performed internally or externally to an organization. Those aspects of software system definition needed to provide the context for software products and services are included. The document also supports the definition, control, assessment, and improvement of these processes. These processes can be applied concurrently, iteratively, and recursively to a software system and its elements throughout the life cycle of a software system. Users of the document can apply the processes and terminology to construct a suitable life cycle model, composed of stages that use selected processes. The process purposes and outcomes in ISO/IEC/IEEE 15288:2023 (systems life cycle processes) and this document are fully aligned so that one may select appropriate activities from either document for use in systems with substantial software content.
8	14764-2021	ISO/IEC/IEEE International Standard - Software engineering - Software life cycle processes – Maintenance Abstract : This standard describes an iterative process for managing and executing software maintenance activities. Use of this standard is not restricted by size, complexity, criticality, or application of the software product. This standard uses a process model to discuss and depict aspects of software maintenance. The criteria established apply to both the planning of maintenance for software while under development, as well as the planning and execution of software maintenance activities for existing software products. Ideally, maintenance planning should begin during planning for software development. This International Standard provides the framework within which generic and specific software maintenance plans may be executed, evaluated, and tailored to the maintenance scope and magnitude of given software products. This International Standard provides the framework, precise terminology, and processes to allow the consistent application of technology (tools, techniques, and methods) to software maintenance. This International Standard provides requirements and guidance for the maintenance of software. The basis for the Maintenance Process and its activities is consistent with ISO/IEC/IEEE 12207:2017, Systems and software engineering -- Software life cycle processes. This standard defines the activities and tasks of software maintenance, and provides maintenance planning requirements. It does not address the operation of software and the operational functions, e.g., backup, recovery, or system administration, which are normally performed by those who operate the software.
9	15026-1-2019	ISO/IEC/IEEE International Standard - Systems and software engineering-- Systems and software assurance --Part 1: Concepts and vocabulary Abstract : This document defines assurance-related terms and establishes an organized set of concepts and their relationships, thereby establishing a basis for shared understanding of the concepts and principles central to all parts of ISO/IEC/IEEE 15026 across its user communities. It provides information to users of the subsequent parts of ISO/IEC/IEEE 15026, including the use of each part and

		the combined use of multiple parts. Coverage of assurance for a service being operated and managed on an ongoing basis is not covered in the ISO/IEC/IEEE 15026 series.
10	15026-2-2022	ISO/IEC/IEEE International Standard - Systems and software engineering-- Systems and software assurance--Part 2: Assurance case Abstract : This document specifies requirements for structure terminology of assurance cases. This document is applicable for developing and maintaining assurance cases.
11	15026-3-2013	IEEE Standard Adoption of ISO/IEC 15026-3 -- Systems and Software Engineering -- Systems and Software Assurance -- Part 3: System Integrity Levels Abstract : The concept of integrity levels with corresponding integrity level requirements that are required to be met in order to show the achievement of the integrity level are specified in this adoption of ISO/IEC 15026-3:2011. It places requirements on and recommends methods for defining and using integrity levels and their integrity level requirements, including the assignment of integrity levels to systems, software products, their elements, and relevant external dependences. This standard is applicable to systems and software and is intended for use by the following: Definers of integrity levels such as industry and professional organizations, standards organizations, and government agencies; Users of integrity levels such as developers and maintainers, suppliers and acquirers, users, and assessors of systems or software and for the administrative and technical support of systems and/or software products. One important use of integrity levels is by suppliers and acquirers in agreements; for example, to aid in assuring safety, economic, or security characteristics of a delivered system or product. This standard does not prescribe a specific set of integrity levels or their integrity level requirements. In addition, it does not prescribe the way in which integrity level use is integrated with the overall system or software engineering life cycle processes. This standard can be used alone or with other parts of ISO/IEC 15026. It can be used with a variety of technical and specialized risk analysis and development approaches. ISO/IEC TR 15026-1 provides additional information and references to aid users of IEEE Std 15026-3.
12	15026-4-2021	ISO/IEC/IEEE International Standard - Systems and software engineering-- Systems and software assurance -- Part 4: Assurance in the life cycle Abstract : This document provides guidance and recommendations for assurance of a selected claim about the system-of-interest by achieving the claim and showing the achievement. The guidance and recommendations are given in a System Assurance process view on top of ISO/IEC/IEEE 15288 and a Software Assurance process view on top of ISO/IEC/IEEE 12207.
13	15288-2023	ISO/IEC/IEEE International Standard - Systems and software engineering-- System life cycle processes Abstract : This document establishes a common framework of process descriptions

		for describing the life cycle of systems created by humans, defining a set of processes and associated terminology from an engineering viewpoint. These processes can be applied to systems of interest, their system elements, and to system of systems. Selected sets of these processes can be applied throughout the stages of a system's life cycle. This is accomplished through the involvement of stakeholders, with the ultimate goal of achieving customer satisfaction.
14	15289-2019	ISO/IEC/IEEE International Standard -- Systems and software engineering - Content of life-cycle information items (documentation) Abstract : The purpose and content of all identified systems and software life cycle and service management information items (documentation) are specified in this document. The information item contents are defined according to generic document types (description, plan, policy, procedure, report, request, and specification) and the specific purpose of the document. This document provides a mapping of ISO/IEC/IEEE 15288:2015 and ISO/IEC/IEEE 12207:2017 clauses with a set of information items. This document identifies records and information items based on analysis of references in ISO/IEC/IEEE 15288:2015 and ISO/IEC/IEEE 12207:2017, which in some cases provide partial or complete outlines for the content of specific documents. Information items may be combined or subdivided as needed for project or organizational purposes.
15	15939-2017	ISO/IEC/IEEE International Standard - Systems and software engineering-- Measurement process Abstract : This document provides an elaboration of the measurement process from ISO/IEC 15288 and ISO/IEC 12207. The measurement process is applicable to system and software engineering and management disciplines. The process is described through a model that defines the activities of the measurement process that are required to adequately specify what measurement information is required, how the measures and analysis results are to be applied, and how to determine if the analysis results are valid. The measurement process is flexible, tailorable, and adaptable to the needs of different users. This document identifies a process that supports defining a suitable set of measures that address specific information needs. It identifies the activities and tasks that are necessary to successfully identify, define, select, apply, and improve measurement within an overall project or organizational measurement structure. It also provides definitions for commonly used measurement terms.
16	16085-2020	ISO/IEC/IEEE International Standard - Systems and software engineering -- Life cycle processes -- Risk management Abstract : This document is an elaboration standard for the risk management process described in ISO/IEC/IEEE 15288 and ISO/IEC/IEEE 12207. This document provides requirements for the tasks and activities of the risk management process in Clause 6, consistent with these life cycle process International Standards. This document provides a definition of the content of the risk management plan and risk treatment plan. This document also provides guidance for how risk management

		outcomes, activities, and tasks pertain to other processes.
17	16326-2019	ISO/IEC/IEEE International Standard - Systems and software engineering - Life cycle processes - Project management Abstract : This document provides normative content specifications for project management plans concerned with systems, and software systems. This document also provides detailed discussion and advice on applying a set of technical management processes that are common to both the system and software life cycles as covered by ISO/IEC/IEEE 15288 and ISO/IEC/IEEE 12207 respectively. The discussion and advice are intended to aid in the preparation of the normative content of project management plans.
18	23026-2023	ISO/IEC/IEEE International Standard - Systems and Software Engineering -- Engineering and Management of Websites for Systems, Software, and Services Information Abstract : This document defines system engineering and management requirements for the life cycle of websites, including strategy, design, engineering, testing and validation, and management and sustainment for intranet and extranet environments. This document applies to those using web technology to present information and communications technology (ICT) information, such as information for users of systems and services, plans and reports for systems and software engineering projects, and documentation of policies, plans, and procedures for IT service management. This document provides requirements for website owners and website providers, managers responsible for establishing guidelines for website development and operations, website engineers, designers, developers, and operations and maintenance staff, who can be external or internal to the website owner's organization. It applies to websites for public access and for limited access, such as for users, customers, and subscribers seeking information on IT systems, products and services. The requirements and recommendations in this document address the following aspects of usability of informational websites and ease of maintenance of managed website operations: a) locating relevant and timely information; b) applying information security management; c) facilitating accessibility and ease of use; d) providing for consistent and efficient development and maintenance practices. This document is not particularly applicable to websites used primarily for marketing or sales, to deliver instructional material (tutorials), or to provide graphical user interfaces (GUI) for business or consumer transactional application processing. However, this document can provide useful insights for managing such sites. This document does not address vendor and product considerations for website engineering and management. This document does not include specifications for application development tools, programming and scripting languages used for websites, metadata tags, or protocols for network communications. It does not address tools or systems used for management or storage of information content (data, documents) that can be presented on websites. This document does not address the design and architecture of software and systems supporting the Internet.

19	24641-2023	ISO/IEC/IEEE International Standard - Systems and Software engineering-- Methods and tools for model-based systems and software engineering Abstract : This International Standard, within the context of methods and tools for MBSSE: • Provides terms and definitions related to MBSSE; • Defines MBSSE-specific processes for model-based systems and software engineering; the processes are described in terms of purpose, inputs, tasks, and outcomes; • Defines methods to support the defined tasks of each process; • Defines tool capabilities to automate/semi-automate tasks or methods. The processes defined in this document are applicable for a single project, as well as for an organization performing multiple projects. These processes are applicable for managing and performing the systems engineering activities based on models within any stage in the life cycle of a system of interest. Annex D (informative) Relationship with other standards describes the relationships between this document and other standards.
20	24748-1-2024	ISO/IEC/IEEE International Standard - Systems and software engineering - Life cycle management - Part 1: Guidelines for life cycle management Scope: This document provides guidance for the life cycle management of systems and software, complementing the processes described in ISO/IEC/IEEE 15288 and ISO/IEC/IEEE 12207. This document:--addresses systems concepts and life cycle concepts, models, stages, processes, process application, key points of view, adaptation and use in various domains and by various disciplines;--establishes a common framework for describing life cycles, including their individual stages, for the management of projects that provide or acquire either products or services;--defines the concept of a life cycle;--supports the use of the life cycle processes within an organization or a project; organizations and projects can use these life cycle concepts when acquiring and supplying either products or services;--provides guidance on adapting a life cycle model and the content associated with a life cycle or a part of a life cycle;--describes the relationship between life cycles and their use in applying the processes... Purpose: The purpose of this document is to facilitate the joint usage of the process content of the latest revisions of both ISO/IEC/IEEE 15288 and ISO/IEC/IEEE 12207, by providing unified and consolidated guidance on life cycle management of systems and software. This is to help ensure consistency in system concepts and life cycle concepts, models, stages, processes, process application, key points of view, adaptation and use in various domains as the two International Standards are used in combination. That will in turn help a project team design a life cycle model for managing the progress of their project. Hence, ISO/IEC/IEEE 15288 and ISO/IEC/IEEE 12207 are the documents that apply the concepts found in this document to specific processes.
21	24748-2-2024	ISO/IEC/IEEE International Standard - Systems and Software Engineering-- Life Cycle Management-- Part 2: Guidelines for the Application of ISO/IEC/IEEE 15288 (System Life Cycle Processes) Scope: This document provides guidance on the application of ISO/IEC/IEEE

		15288. It addresses the application of system, life cycle, organizational, project, process, and conformance and adaption concepts, principally through references to ISO/IEC/IEEE 24748-1 and ISO/IEC/IEEE 15288. This document gives guidance on applying ISO/IEC/IEEE 15288 from the aspects of strategy, planning, application in organizations and application on projects. It also provides a comparison of the differences between ISO/IEC/IEEE 15288:2023 and ISO/IEC/IEEE 15288:2015.
22	24748-3-2020	ISO/IEC/IEEE International Standard - Systems and software engineering-- Life cycle management--Part 3: Guidelines for the application of ISO/IEC/IEEE 12207 (software life cycle processes) Abstract : ISO/IEC/IEEE 24748-3 provides guidance on the application of the software life cycle processes standard, ISO/IEC/IEEE 12207:2017. This document establishes guidance to implement a common framework for software life cycle processes, with well-defined terminology, that can be referenced by the software industry. This document provides guidance on defining, controlling, and improving software life cycle processes within an organization or a project. It recommends methods and approaches suitable for a variety of life cycle models. The guidance emphasizes the importance of establishing a strategy, planning, and the involvement of stakeholders, with the ultimate goal of achieving customer satisfaction. Its purpose is to help ensure consistency in system concepts and life cycle concepts, models, stages, processes, process application, key points of view, adaptation and use in various domains. This document concentrates on specific guidance for the Technical processes and how they can be effectively used during the software life cycle. It is intended to be useful in a variety of software life cycle situations, including the use of agile methods.
23	24748-4-2025	ISO/IEC/IEEE International Standard - Systems and Software Engineering -- Life Cycle Management -- Part 4: Systems Engineering Management Planning Abstract : ISO/IEC/IEEE 15288, Systems and software engineering – System life cycle processes, provides a common process framework covering the life cycle of human-made systems, from the conception of ideas through to the retirement of a system. It provides the processes for acquiring and supplying systems. In addition, this framework provides for the assessment and improvement of the life cycle processes. This common framework improves communication and cooperation among the parties that create, utilize, and manage modern systems in order that they can work in an integrated, coherent fashion. The acquisition or supply of a system is usually done within a project. A project prepares and implements the technical planning, resources and schedules necessary to guide the project toward accomplishment of its objectives and proper conclusion. The project's authorization and objectives are documented in an information item identified as a Systems Engineering Management Plan (SEMP). This document defines...
24	24748-6-2023	ISO/IEC/IEEE International Standard - Systems and Software Engineering -- Life Cycle Management -- Part 6: Systems and Software Integration Abstract : This document: --provides supplemental requirements and guidance for

		the planning and performing of the integration processes given in ISO/IEC/IEEE 15288 and ISO/IEC/IEEE 12207; --provides guidance on the relationship between the integration process and other life cycle processes. --specifies requirements for information items to be produced as a result of using the integration process, including the content of the information items. This document is applicable to: -- those who use or plan to use ISO/IEC/IEEE 12207 or ISO/IEC/IEEE 15288, or both, on projects dealing with human-made systems, software-intensive systems, and products and services related to those systems, regardless of the project scope, methodology, size, or complexity; --anyone planning or performing integration activities to aid in ensuring that the application of the integration process and its relationships to other system life cycle processes conform to ISO/IEC/IEEE 15288 or ISO/IEC/IEEE 12207.
25	24748-7-2025	ISO/IEC/IEEE International Standard - Systems and software engineering-- Life cycle management--Part 7: Application of systems engineering on defense programs Abstract : This document implements ISO/IEC/IEEE 15288 for application on defence programs, providing the defence-specific language and terminology to help ensure the correct application of acquirer-supplier requirements for a defence program. It provides the basis for selection, negotiation, agreement, and performance of necessary systems engineering activities and delivery of products, while allowing flexibility for both innovative implementation and tailoring of the specific systems engineering process(es) to be used by system suppliers, either contractors or government system developers, integrators, maintainers, or sustainers. This document includes the expected or required outputs and associated attributes.
26	24748-9-2023	ISO/IEC/IEEE International Standard - Systems and software engineering -- Life cycle management -- Part 9: Application of system and software life cycle processes in epidemic prevention and control systems Abstract : Many countries have adopted information-based prevention and control measures during the epidemic, and have developed numerous epidemic prevention and control systems and software. Most of the processes in the entire life cycle of the epidemic prevention and control systems are most likely to be completed in the event of an epidemic. Compared with the normal state, there will be special situations such as poor communication caused by the need for personnel to maintain a safe distance; limited transportation and logistics, resulting in insufficient infrastructure protection; short delivery cycles and frequent iterative upgrades; and special requirements such as accuracy, disaster tolerance, degradation capability, safety, user capacity and stress testing, and rapid demand capture. In this case, it is believed that in the development process of epidemic prevention and control systems, the application of the life cycle process method specified in the ISO/IEC/IEEE 15288:2015 or ISO/IEC/IEEE 12207 standards can effectively help guide the process management and application of the epidemic prevention and control systems.

27	24748-10-2025	ISO/IEC/IEEE International Standard - Systems and software engineering--Life cycle management--Part 10: Guidelines for systems engineering agility Abstract : Systems engineering agility is a strategy-based method for designing, building, sustaining, and evolving purpose-fulfilling creations when knowledge is uncertain and operational environments are dynamic. Strategies are abstractions for what needs to be accomplished and why, without constraints or directions on how to achieve them. This part of ISO/IEC/IEEE 24748 specifies strategic aspects supporting systems engineering agility. It includes guidelines for the application of the strategic aspects as well as relevant examples. Individually and collectively, the aspects can improve capability to deal with uncertain knowledge and dynamic environments.
28	24774-2021	ISO/IEC/IEEE International Standard -Systems and software engineering - Life cycle management--Specification for process description Abstract : A number of international, national and industry standards describe processes and process models. Process models are developed for a range of purposes including process implementation and assessment. In order for users of standards to select the appropriate forms of process description and apply them in a consistent fashion, it is desirable to develop a common characterization of all of these forms of process description. This document presents requirements for the description of processes in terms of their format, content and level of prescription. Because of the wide usage of process descriptions, process elements are categorized as “Required” or “Optional”. Although developed within ISO/IEC/JTC1/SC7 - Systems and software engineering, the requirements of this document can be applied to any process model or description developed for any purpose.
29	26511-2018	ISO/IEC/IEEE International Standard - Systems and software engineering - Requirements for managers of information for users of systems, software, and services Abstract : This document specifies requirements and procedures for managing information for users throughout the software-, services-, and systems-development life cycle. It applies to people or organizations producing suites of information, to those undertaking a single information-development project, and to information produced internally, as well as to information contracted to outside service organizations. It provides details of the information-development and project-management processes and also presents aspects of strategic planning and translation management that managers apply. It covers management activities related to starting a project, building a team, and managing information through the translation and localization process. It addresses productivity and quality measurements needed for management control. This document is independent of the software tools that may be used to produce or manage information for users and applies to both printed, embedded, and mobile information. Much of its guidance is applicable to information for users of systems of hardware as well as software, systems, and services.

30	26512-2026	ISO/IEC/IEEE International Standard - Systems and software engineering -- Requirements for acquirers and suppliers of information products and services Abstract : This document was developed to assist users of ISO/IEC/IEEE 15288 or ISO/IEC/IEEE 12207 to acquire or supply information for users as part of the system or life cycle processes. It defines the documentation process from the acquirer's standpoint and the supplier's standpoint. This document covers the requirements for information items used in the acquisition of user documentation products: the acquisition plan, information product specification, statement of work, request for proposals, and the proposal. It provides an overview of the information management processes which may require acquisition and supply of system or software user documentation products and services. It addresses the preparation of requirements for user documentation. These requirements are central to the user documentation specification and statement of work. It includes requirements for primary document outputs of the acquisition and supply process: the request for proposal and the proposal for user documentation ...
31	26513-2017	ISO/IEC/IEEE International Standard - Systems and software engineering -- Requirements for testers and reviewers of information for users Abstract : This document supports the need for software users to receive consistent, complete, accurate, and usable documentation. This document was developed to assist those who review, assess, and test software user documentation as part of the software lifecycle process. It is not limited to the review and test phase of the software lifecycle, but includes activities throughout the information management and documentation management processes. This document applies to printed and electronic documentation such as user manuals, online help, instructions, user assistance, tutorials, websites, and user reference documentation. Much of its guidance is applicable to user documentation for systems as well as the software used to control machinery or hardware devices. It applies to both initial development and subsequent releases of the software and user documentation. This document is intended for use by all types of organizations, whether or not a dedicated documentation department is present. Readers are assumed to have experience with or general knowledge of reviewing and testing processes.
32	26514-2021	ISO/IEC/IEEE International Standard - Systems and software engineering -- Design and development of information for users Abstract : This document covers the development process for designers and developers of information for users of software. It describes how to establish what information users need, how to determine the way in which that information should be presented, and how to prepare the information and make it available. It is not limited to the design and development stage of the life cycle, but includes information on design throughout the life cycle, such as design strategy and maintaining a design. This document provides requirements for the structure, information content, and format of information for users of software. This document can be applied to developing the following types of information, although it does not

		cover all aspects of them: - information for users of products other than software; - multimedia systems using animation, video, and sound; - computer-based training (CBT) packages and specialized course materials intended primarily for use in formal training programs; - maintenance information describing the internal operation of systems software; - information for users incorporated into the user interface itself. This document is applicable to information architects and information developers, including a variety of specialists: - information architects who plan the structure and format of information products; - usability specialists and business analysts who identify the tasks that the intended users can perform with the software; - developers and editors of the written content of information for users; - graphic designers with expertise in electronic media; - user interface designers and ergonomics experts working together to design the presentation of the information on the screen. This document is also a reference for those with other roles and interests in the process of developing information for users: - managers of the software development process or the information-development process; - acquirers of information for users prepared by suppliers; - usability testers, reviewers of information for users, subject-matter experts; - developers of tools for creating information for users; - human-factors experts who identify principles for making information for users more accessible and easily used.
33	26515-2018	ISO/IEC/IEEE International Standard - Systems and software engineering — Developing information for users in an agile environment Abstract : This document provides requirements and guidance to information developers and related roles on how to adapt the processes described in the ISO/IEC/IEEE 265NN family of International Standards to develop quality information for users. This document is independent of the agile development methods and tools that are used to produce the software. This document gives an overview of agile methodologies although it neither encourages nor discourages the use of any particular agile methodology. Therefore, this document uses generic agile terminology as much as possible.
34	26516-2025	ISO/IEC/IEEE International Standard - Systems and software engineering -- Development and production of instructional videos Abstract : This document provides requirements and guidance for the effective planning, design, and development of instructional video for information technology systems, products and services. Videos can present instructions for installation, operation, maintenance, and disposal for skilled or unskilled users. This document includes the use of video, interactive or hypervideo, animation, and virtual or augmented reality to explain how to use software and systems including hardware. This document covers both short video segments and the use of video in larger information products. Instructional videos can include concepts, instructional steps, and reference material. This document specifies content elements, structure, and use of media, including music, narration, captions, titles, and graphics. This document is applicable to instructional video developers, including a variety of specialists such

		as: information architects who plan the structure and format of information products, user experience...
35	26531-2023	ISO/IEC/IEEE International Standard - Systems and software engineering-- Content management for product life cycle, user and service management information for users Abstract : This document provides requirements for the management of the content used in product life cycle, software, and service management system information for users. Content management allows an organization to control the storage and retrieval of content objects, track content revisions, maintain a content audit trail, and enable a collaborative environment. Component content management supports multiple deliverable formats and the reuse of content objects among deliverables. Content objects that are unique and are maintained as independent database objects are efficient to review, approve, and update; may be combined to produce multiple deliverables; and are cost-effective to translate. This document defines the characteristics of an effective and efficient process through which content is gathered, managed, and published, including the requirements of a system that is supported by an electronic database. Such a database should support documents or topics and content units that may be ass...
36	29119-1-2021	ISO/IEC/IEEE International Standard - Software and systems engineering -- Software testing --Part 1: General concepts Abstract : This document specifies general concepts in software testing and presents key concepts for the ISO/IEC/IEEE 29119 series.
37	29119-2-2021	ISO/IEC/IEEE International Standard - Software and systems engineering - Software testing -- Part 2: Test processes Abstract : This document specifies test processes that can be used to govern, manage and implement software testing for any organization, project or testing activity. It comprises generic test process descriptions that define the software testing processes. Supporting informative diagrams describing the processes are also provided. This document is applicable to testing in all software development lifecycle models. This document is intended for, but not limited to, testers, test managers, developers and project managers, particularly those responsible for governing, managing and implementing software testing.
38	29119-3-2021	IEEE/ISO/IEC International Standard for Software and systems engineering-- Software testing--Part 3: Test documentation Abstract : The purpose of the ISO/IEC/IEEE 29119 series of software testing standards is to define an internationally-agreed set of standards for software testing that can be used by any organization when performing any form of software testing. ISO/IEC/IEEE 29119-3 includes templates and examples of test documentation. The templates are arranged within clauses reflecting the overall test process description structure in ISO/IEC/IEEE 29119-2, i.e. by the test process in which they are being produced. Annex A outlines the contents of each document and maps each document to ISO/IEC/IEEE 29119-2. Annex B contains an overview of the examples. Annexes

		C to R contain examples of the application of the templates. Annex S provides mappings to existing standards. Annex T explains the use of the term Test Model in this document. The Bibliography for this part of ISO/IEC/IEEE 29119 is at the end of the document. ISO/IEC/IEEE 29119-3 supports dynamic testing, functional and non-functional testing, manual and automated testing, and scripted and unscripted testing. The documentation templates defined in ISO/IEC/IEEE 29119-3 can be used in conjunction with any software development lifecycle model.
39	29119-4-2021	IEEE/ISO/IEC International Standard - Software and systems engineering-- Software testing--Part 4: Test techniques Abstract : The purpose of this part of ISO/IEC/IEEE 29119 is to provide an International Standard that defines software test design techniques (also known as test case design techniques or test methods) that can be used within the test design and implementation process that is defined in ISO/IEC/IEEE 29119-2. This part of ISO/IEC/IEEE 29119 does not describe a process for test design and implementation; instead, it describes a set of techniques that can be used within the test design and implementation process defined in ISO/IEC/IEEE 29119-2. The intent is to describe a series of techniques that have wide acceptance in the software testing industry. The test design techniques presented in this part of ISO/IEC/IEEE 29119 can be used to derive test cases that, when executed, generate evidence that test item requirements have been met or that defects are present in a test item (i.e. that requirements have not been met). Risk-based testing could be used to determine the set of techniques that are applicable in specific (risk-based testing is covered in ISO/IEC/IEEE 29119-1 and ISO/IEC/IEEE 29119-2).
40	29148-2018	ISO/IEC/IEEE International Standard - Systems and software engineering -- Life cycle processes -- Requirements engineering Abstract : This document contains provisions for the processes and products related to the engineering of requirements for systems and software products and services throughout the life cycle. It defines the construct of a good requirement, provides attributes and characteristics of requirements, and discusses the iterative and recursive application of requirements processes throughout the life cycle. This document provides additional guidance in the application of requirements engineering and management processes for requirements-related activities in ISO/IEC/IEEE 12207 and ISO/IEC/IEEE 15288. Information items applicable to the engineering of requirements and their content are defined. The content of this document can be added to the existing set of requirements-related life cycle processes defined by ISO/IEC/IEEE 12207 or ISO/IEC/IEEE 15288, or can be used independently.
41	32430-2024	ISO/IEC/IEEE International Standard Software engineering - Software Non-Functional Sizing Measurement Abstract : This document defines a method for measuring the non-functional size of the software. It complements ISO/IEC 20926:2009, which defines a method for measuring the functional size of the software. This document also describes the

		complementarity of functional and non-functional sizes, so that deriving the sizes from the functional and the non-functional requirements does not result in duplication in the distinct functional and non-functional sizes. In general, there are many types of non-functional requirements. Moreover, non-functional requirements and their classification evolve over time as the technology advances. This document does not intend to define the type of NFR for a given context. Users can choose ISO 25010 or any other standard for the definition of NFR. It is assumed that users size the NFR based on the definitions they use. This document covers a subset of non-functional requirements. It is expected that, with time, the state of the art can improve and that potential future versions of this document can define an extended coverage. The ultimate goal is a version that, together with ISO/IEC 20926:2009, covers every aspect that can be required of any prospective piece of software, including aspects such as process and project directives that are hard or impossible to trace to the software's algorithm or data. The combination of functional and non-functional sizes would then correspond to the total size necessary to bring the software into existence. Estimating the cost, effort and duration of the implementation of the NFR is outside the scope of this document.
42	32675-2021	ISO/IEC/IEEE International Standard--Information technology--DevOps--Building reliable and secure systems including application build, package and deployment Abstract: Technical principles and processes to build, package, and deploy systems and applications in a reliable and secure way are specified. Establishing effective compliance and information technology (IT) controls is the focus. DevOps principles presented include mission first, customer focus, left-shift, continuous everything, and systems thinking. How stakeholders, including developers and operations staff, can collaborate and communicate effectively is described. The process outcomes and activities herein are aligned with the process model specified in ISO/IEC/IEEE 12207:2017 and ISO/IEC/IEEE 15288:2015.
43	42010-2022	IEEE/ISO/IEC International Standard for Software, systems and enterprise--Architecture description Abstract : This document specifies requirements for the structure and expression of an architecture description (AD) for various entities, including software, systems, enterprises, systems of systems, families of systems, products (goods or services), product lines, service lines, technologies and business domains. This document distinguishes the architecture of an entity of interest from an AD expressing that architecture. Architectures are not the subject of this document. This document specifies requirements for use of the architectural concepts and their relationships as captured in an AD. It does not specify requirements for any entity of interest or its environment. This document specifies requirements for an architecture description framework (ADF), an architecture description language (ADL), architecture viewpoints and model kinds in order to usefully support the development and use of an AD. This document specifies conformance to the requirements for an AD, ADF, ADL, architecture viewpoint and model kind. This document does not specify the

		processes, architecting methods, models, notations, techniques or tools by which an AD is created, utilized or managed. This document does not specify any format or media for recording an AD.
44	90003-2018	ISO/IEC/IEEE International Standard - Software engineering -- Guidelines for the application of ISO 9001:2015 to computer software Abstract : This document provides guidance for organizations in the application of ISO 9001:2015 to the acquisition, supply, development, operation and maintenance of computer software and related support services. This document does not add to or otherwise change the requirements of ISO 9001:2015. The guidelines provided in this document are not intended to be used as assessment criteria in quality management system registration/certification. The application of this document is appropriate to software that is: part of a commercial contract with another organization, a product available for a market sector, used to support the processes of an organization, embedded in a hardware product, or related to software services. Some organizations may be involved in all the above activities; others may specialize in one area. Whatever the situation, the organization's quality management system should cover all aspects (software related and non-software related) of the business. This document identifies t...
45	3110-2025	IEEE Standard for Computer Vision (CV)--Technical Requirements for Algorithms Application Programming Interfaces (APIs) of Deep Learning Framework Abstract : Functional and technical requirements for the interfaces between algorithms and learning frameworks (including the interfaces provided by training frameworks), and between algorithms and data sets in the development of artificial intelligence (AI) computer vision algorithms are specified in this standard. The adaptation and invocation of the application programming interfaces (API) involved in the development of computer vision algorithms (CVA) based on deep learning (hereinafter referred to as “CVA”) is addressed in this standard
46	3358-2025	IEEE Standard for Technical Requirements for Low-Code Application Platform Abstract : The architectural and functional requirements of the low-code development platform are specified, as are the performance requirements such as scalability, portability, and security. It is applicable to providing guidance and specifications for the design, implementation, and application of low-code development platforms.
47	3079.3.1-2025	IEEE Standard for Service Application Programming Interfaces (APIs) for Digital Human Authoring and Visualization Abstract : Six service application programming interfaces (APIs) and a framework required for digital human authoring and visualization services are defined in this standard. A system architecture and API interfaces are included in this framework.

48	3407-2025	IEEE Standard for End-to-End Software Testing Automation Tools Abstract : The minimum set of requirements for end-to-end software testing automation tools is established in this standard. Also, this standard can be used to guide users and industries to prepare and perform different types of automated testing on software integration environments. These tools can help users and industries in streamlining the software test automation processes, leading to a reduction in cost and effort on test development, execution, and maintenance.
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